

Project Deliverable 1

The system

The system is a solar oven that heats a pipe that transfers hot nitrogen for a reactor that is doing pyrolysis. Its boundary is defined by the interfaces it has with the system it is coupled with. The system is coupled with two subsystems that define its boundary. One of the couplings comes from a nitrogen generator and another one comes from the pyrolysis reactor itself. With respect to the environment, solar energy is considered an input to the system along with the room temperature nitrogen and occasional handling of the personnel for maintenance, the output of the system is the heated nitrogen flow. Thus, the system boundary is characterized by the definition between the expected inputs (sunlight, room temperature nitrogen flow and personnel handling) and outputs (heated nitrogen flow).

The stakeholders

- Nitrogen generator subsystem managers
- Pyrolysis reactor subsystem managers
- Maintenance operators of the system
- Plant managers and owners
- Industrial safety and compliance regulators

The questions

- How the components react to changes in the input?
- How does the reactor capacity reacts to changes in the sunlight?
- Are the safety requirements from the input traced to a verification strategy?
- Are the safety requirements from the output traced to a verification strategy?
- Does the architecture offer a way of defining modular replacement of the components?

Why it is hard?

- Knowledge is fragmented because there are different specialists needed to implement this system. Namely, optics specialists that can understand how light travels and gets refracted or reflected, mechanical engineering knowledge that will explain the modular way of assembling the system, chemical engineering

knowledge that will guarantee chemical safety and innocuity a heated flow over the system.

- The system works in at least two flow levels making the concerns more coupled and intricate, there is the nitrogen flow level (matter level) and the energy level (sunlight to heated pipe and heated gas flow).

Available data

- The data for this system will be realized through simulations and design descriptions, the system will not be implemented in the real world but tested over different simulations to show its validity.
- The data used to generate the simulations will come from research over the internet, public sources and engineering textbooks.